

CENTRO FEDERAL DE EDUCAÇÃO TECNOLÓGICA DE MINAS GERAIS
DEPARTAMENTO DE FÍSICA

Prof. Kilder Leite Ribeiro

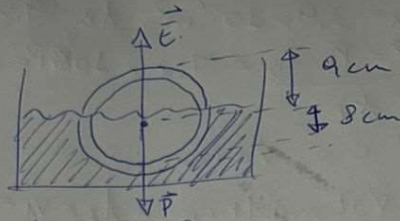
Aluno: **Enzo Rocha Leite Diniz Ribas**

Fundamento de Ondas Fluidos e Termodinâmica Segundo Teste

28/03 2026 Enzo R. L. D. Ribos Teste II OFT

Q1 Esfera Oca

$$\begin{cases} r_{\text{ext}} = 9,0\text{cm} = (R) \\ r_{\text{int}} = 8,0\text{cm} \\ \rho_{\text{liq}} = 800\text{kgm}^{-3} \end{cases}$$



a) Qual é a massa da esfera?

$$\sum \vec{F} = 0 \Rightarrow \vec{P} + \vec{E} = 0 \quad (\uparrow +) \Rightarrow E = P$$

$$|mg = \rho_{\text{liq}} V_{\text{sub}} \cdot g| \quad V = \frac{4}{3}\pi R^3 \quad V_{\text{sub}} = \frac{1}{2} V = \frac{2}{3}\pi R^3$$

$$\Rightarrow m g = \rho_{\text{liq}} \frac{2}{3}\pi R^3 g \quad m = \rho_{\text{liq}} \frac{2}{3}\pi R^3$$

$$m = 800 \frac{\text{kg}}{\text{m}^3} \left(\frac{2}{3}\right) \pi (0,09\text{m})^3$$

$$= \frac{800\pi(2)}{3} (0,09)^3 \text{ kg} \approx 1,22 \text{ kg}$$

b) Calcule a massa específica do que é feita a esfera.

$$\rho_{\text{est}} = \frac{M}{V} \quad V = V_{\text{ext}} - V_{\text{int}} \\ \Rightarrow V = \frac{4}{3}\pi (R^3 - r_{\text{int}}^3)$$

$$\rho_{\text{est}} = \frac{1,22 \text{ kg} (3)}{4\pi [(0,09)^3 - (0,08)^3] \text{ m}^3} = \frac{1,22 \text{ kg} (3)}{4\pi (2,17 \times 10^{-4}) \text{ m}^3}$$

$$\rho_{\text{est}} = 1342,18 \frac{\text{kg}}{\text{m}^3}$$

$$Q2 \left\{ \begin{array}{l} \rho_{\text{liq}} = 900 \text{ kg m}^{-3}; \\ A_A = 1,9 \times 10^{-2} \text{ m}^2; A_B = 9,5 \times 10^{-2} \text{ m}^2; \\ \Delta P = P_A - P_B = 7,2 \times 10^3 \text{ Pa}; h_A = h_B \quad (0.3) \end{array} \right.$$

(★)

Q) Qual é a vazão

$$\text{Vazão Volumétrica} = A_A \times V_A = A_B \times V_B = Q$$

$$V_A = \frac{A_B}{A_A} V_B \quad \frac{A_B}{A_A} = \frac{9,5 \times 10^{-2} \text{ m}^2}{1,9 \times 10^{-2} \text{ m}^2} = 5$$

$$\Rightarrow V_A = 5 V_B \quad (1)$$

$$P_A + \frac{1}{2} \rho V_A^2 = P_B + \frac{1}{2} \rho V_B^2$$

$$P_A - P_B = \frac{1}{2} \rho (V_B^2 - V_A^2) \quad \text{Recolli: (1) e } \star$$

$$\Delta P = \frac{1}{2} \rho (5 V_B^2 - V_B^2) \quad \Delta P = \frac{1}{2} \rho (25 V_B^2 - V_B^2)$$

$$= \frac{1}{2} \rho (24 V_B^2) = \rho (12) V_B^2$$

$$\Rightarrow V_B = \sqrt{\frac{\Delta P}{\rho (12)}} = \sqrt{\frac{7,2 \times 10^3 \text{ Pa}}{12 (900 \frac{\text{kg}}{\text{m}^3})}}$$

$$= \sqrt{0,67} \sqrt{\frac{\text{kg m/s}^2}{\text{m}^2} \cdot \frac{\text{m}^3}{\text{kg}}} = \sqrt{0,67 \frac{\text{m}^2}{\text{s}^2}} = 0,82 \frac{\text{m}}{\text{s}}$$

$$Q = A_B \times V_B = (9,5 \times 10^{-2} \text{ m}^2) (0,82 \frac{\text{m}}{\text{s}}) \approx 7,69 \times 10^{-2} \text{ m}^3/\text{s}$$

b) Qual é a vazão mássica?

$$\dot{m} = \rho \times Q = \frac{900 \text{ Kg}}{\text{m}^3} \times 7,69 \times 10^{-2} \frac{\text{m}^3}{\text{s}}$$
$$= 69,25 \text{ Kg/s}$$